5T+1 CFET SRAM with Dual-Bitline-per-Cell-Height Enabling Differential Readout and 12.5% Area Reduction

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Outline

- CFET SRAM Scaling Challenges
  - 6T CFET SRAM → underutilized CFET stack
  - 5T CFET SRAM → writability limitation + single-ended sensing
- Proposed 5T+1 CFET SRAM
  - Bitcell design and layout implementation
  - Dual-bitline-per-cell-height → differential sensing recovery
  - Split-V_{DD} → writability improvement
- DTCO-based validation and results
- Conclusion

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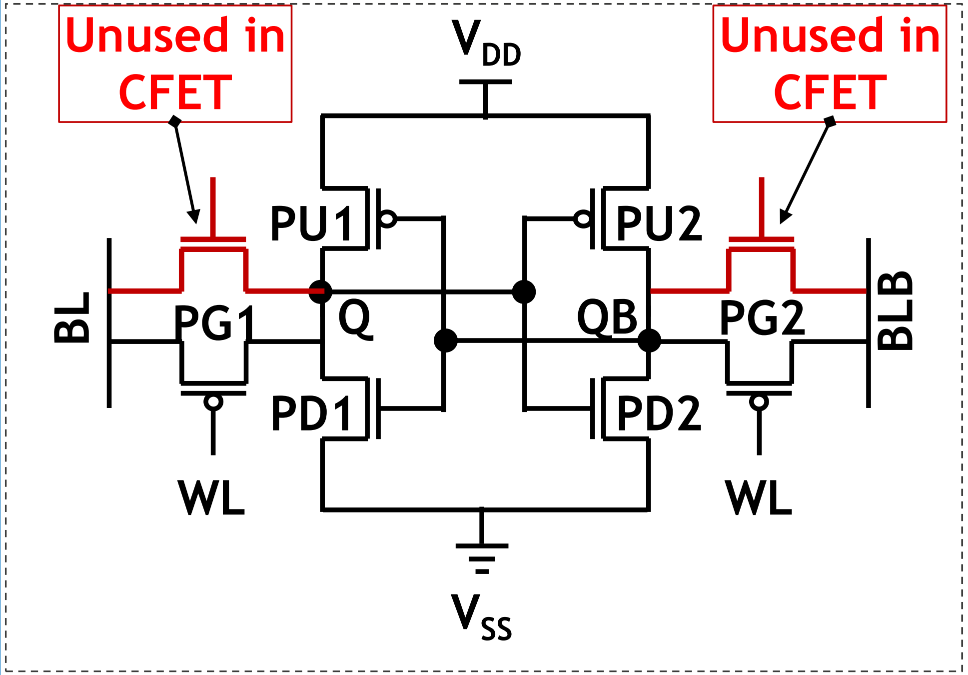
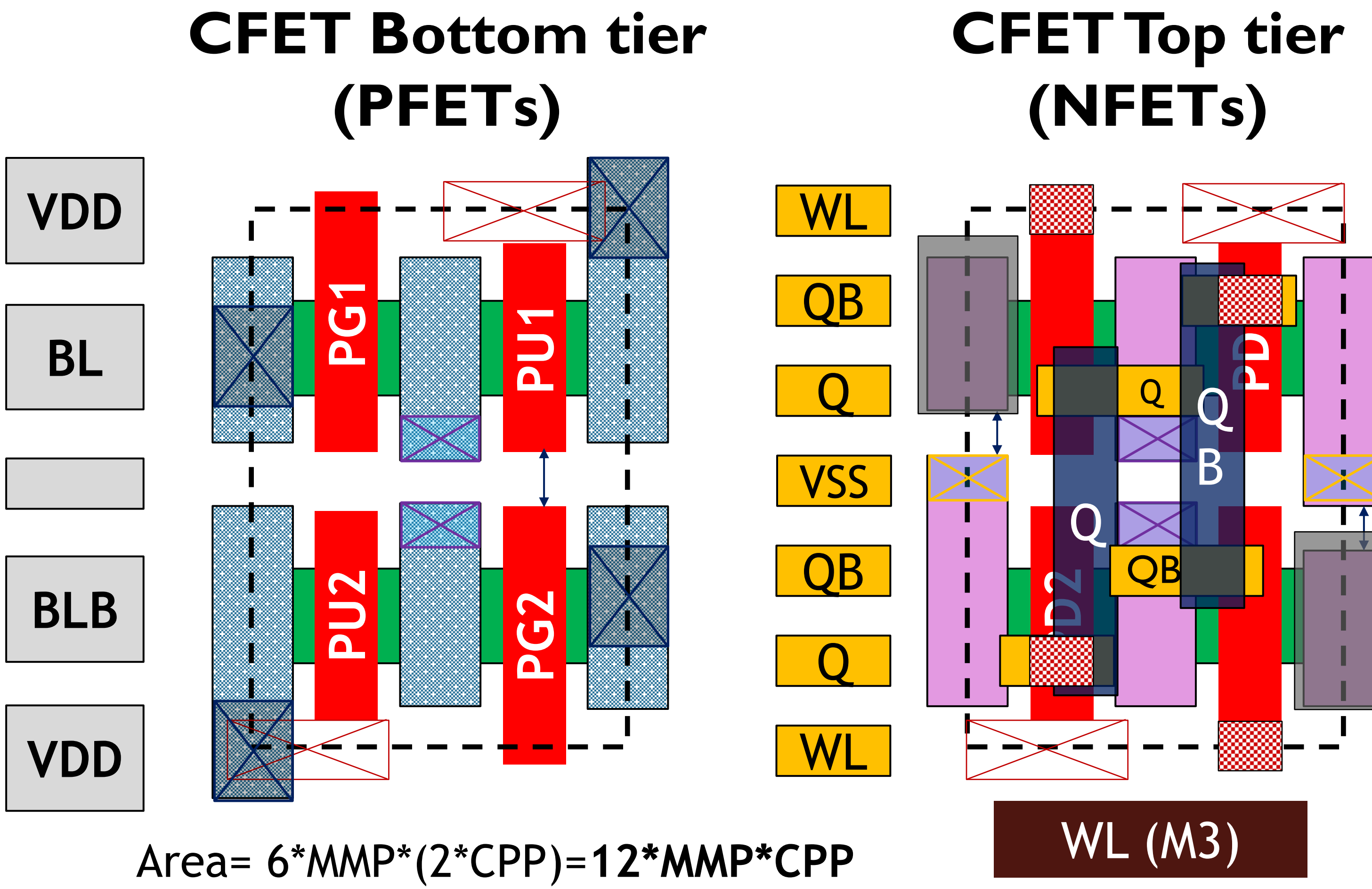
6T CFET SRAM: underutilized vertical stack

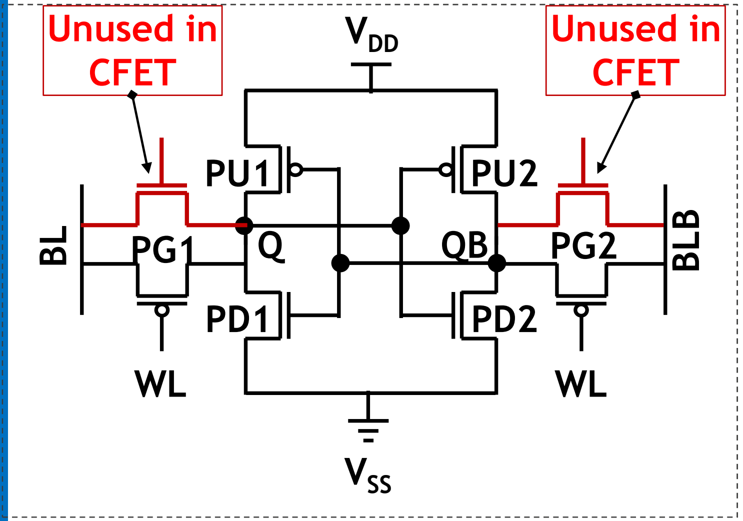




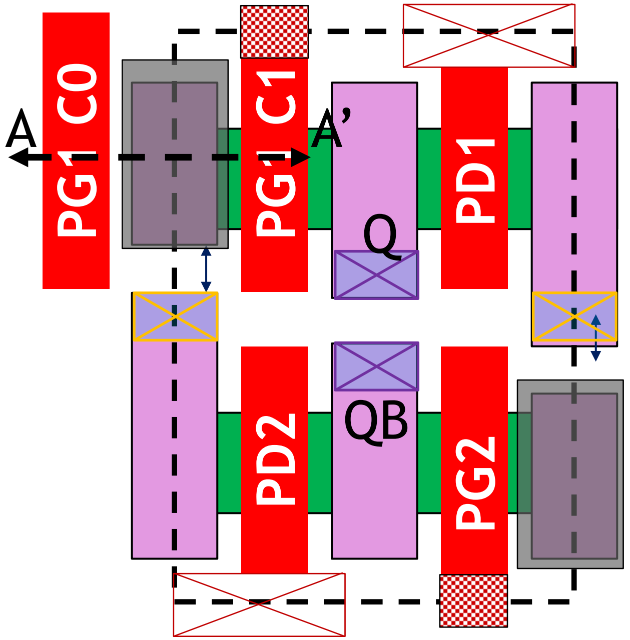
Area= 6*MMP*(2*CPP)=**12*MMP*CPP**

Unused stacked devices → CFET density not fully utilized

6T CFET SRAM requires nanosheet cut

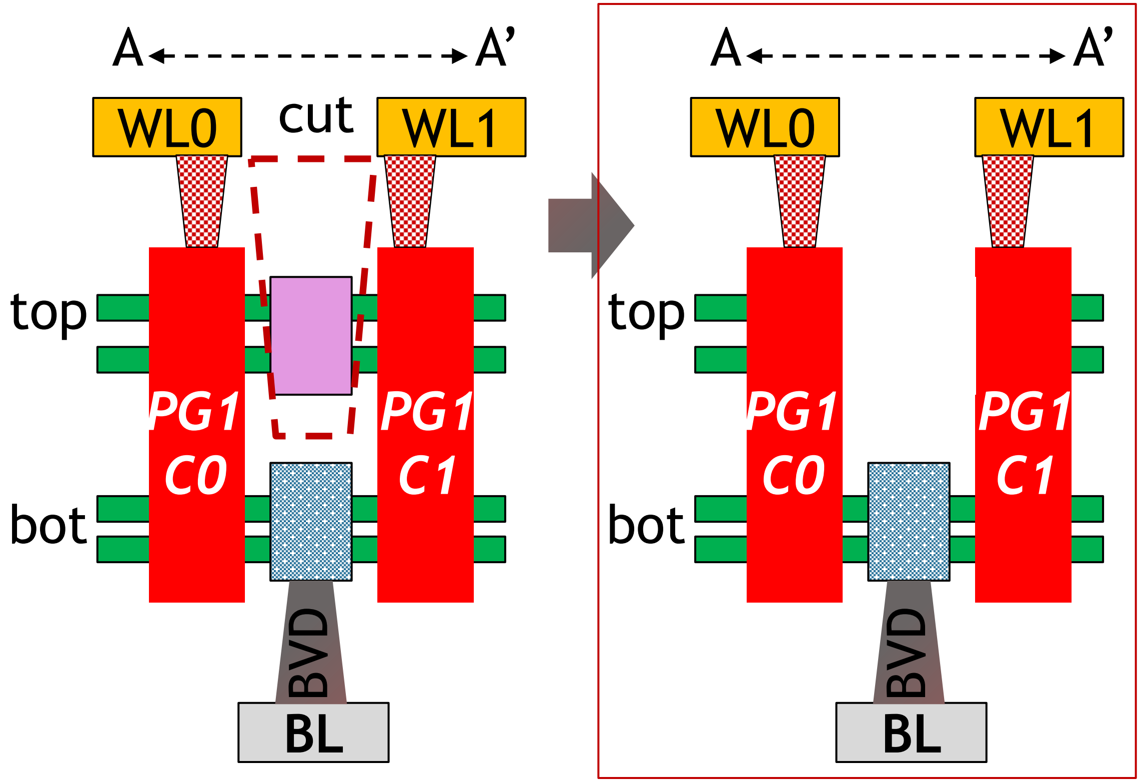


CFET Top tier (NFETs)



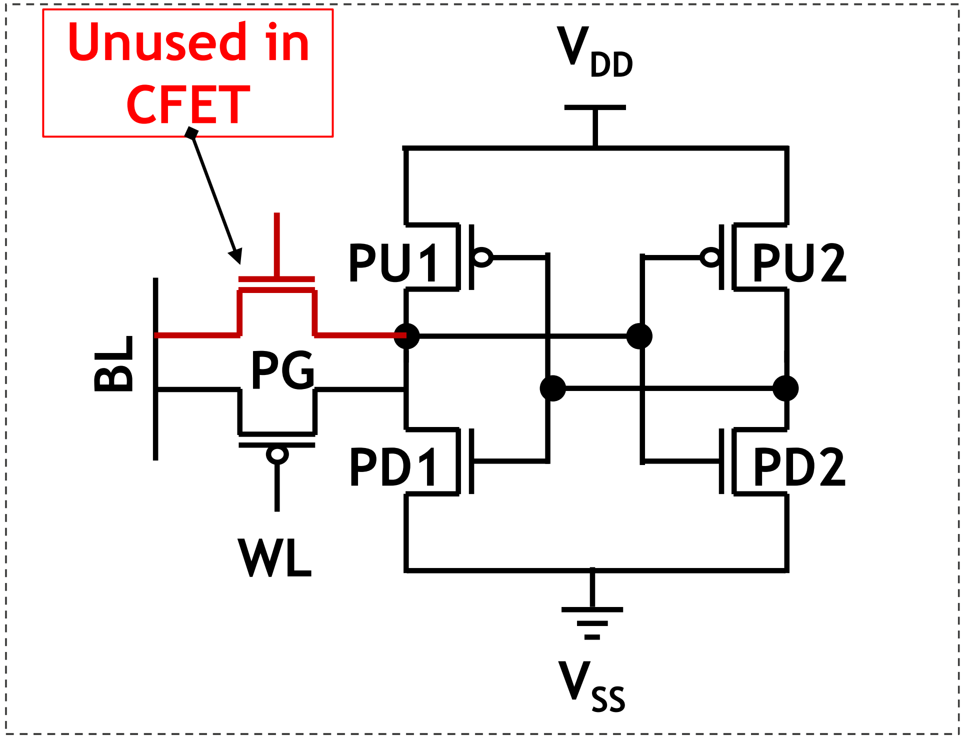
Area= 6*MMP*(2*CPP)=**12*MMP*CPP**

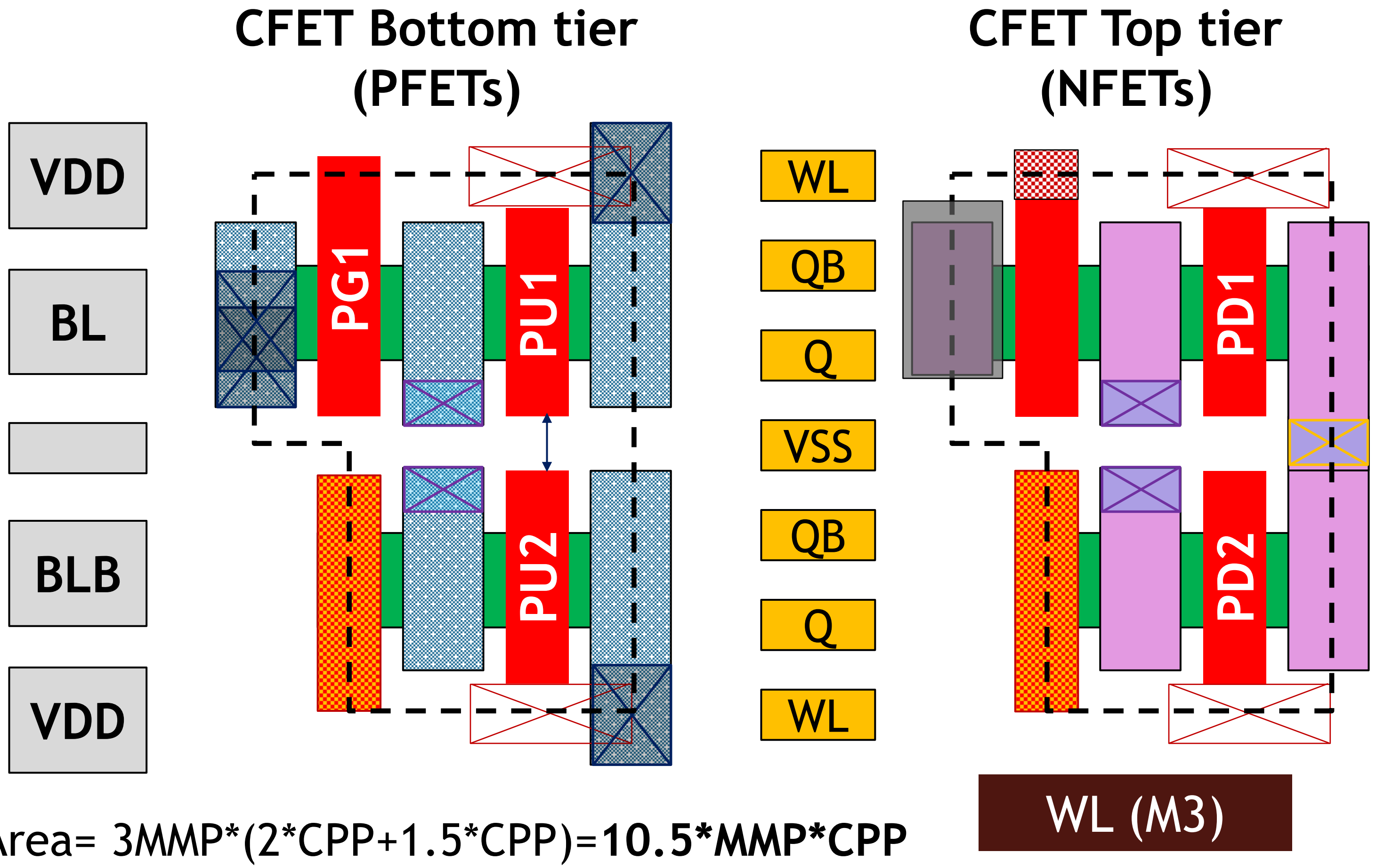
Top nanosheet cut



Unused devices → nanosheet cut → increased process complexity

5T CFET SRAM: reduced footprint vs 6T

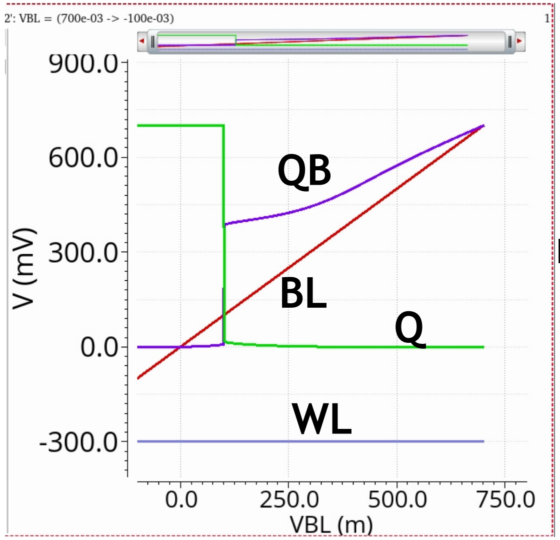


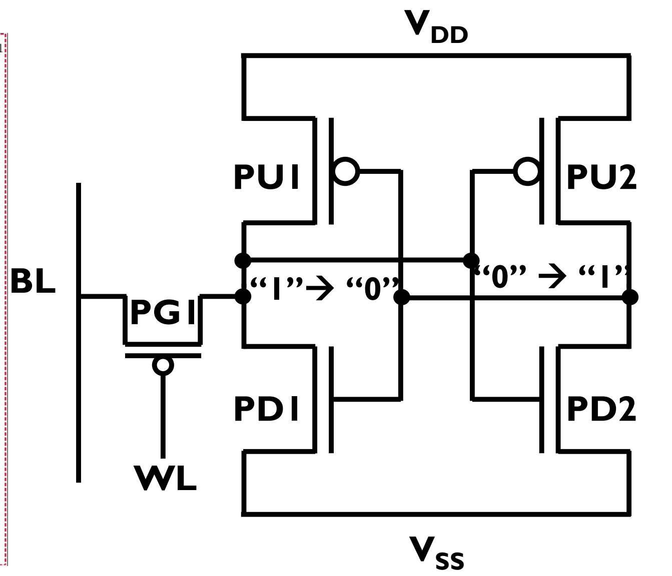


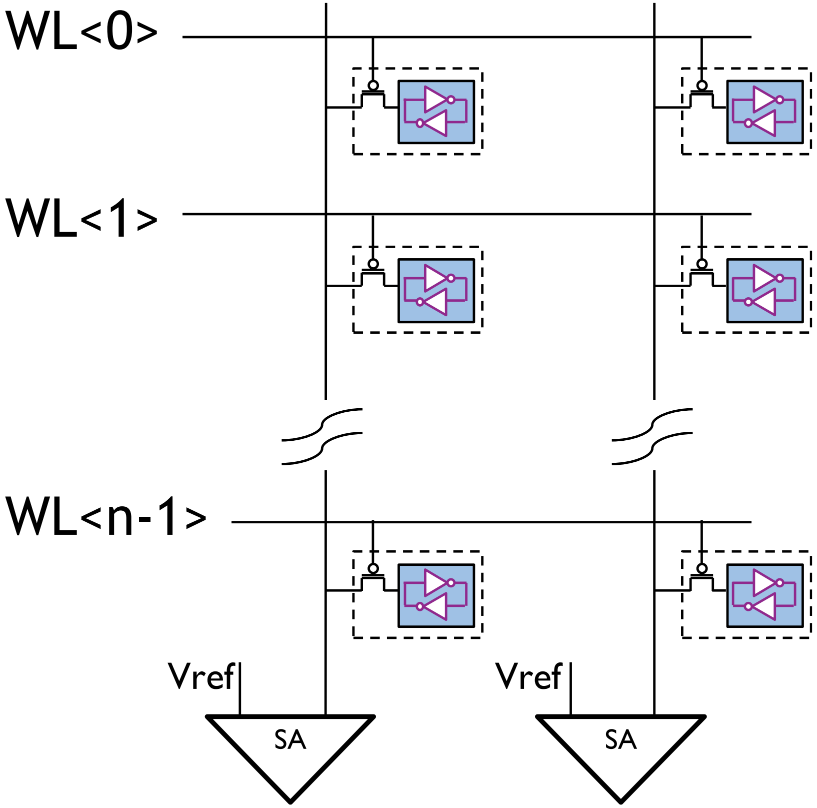
Area= 3MMP*(2*CPP+1.5*CPP)=**10.5*MMP*CPP**

-12.5% area vs 6T; effective width reduction (non-rectangular)

5T CFET SRAM: write limitation and single-ended read sensing penalty







- Require WL underdrive (~-300mV)
- Write '0' fails without strong assist

- Larger BL swing →slower read
- Limits bitline length (~64 cells/bitline)

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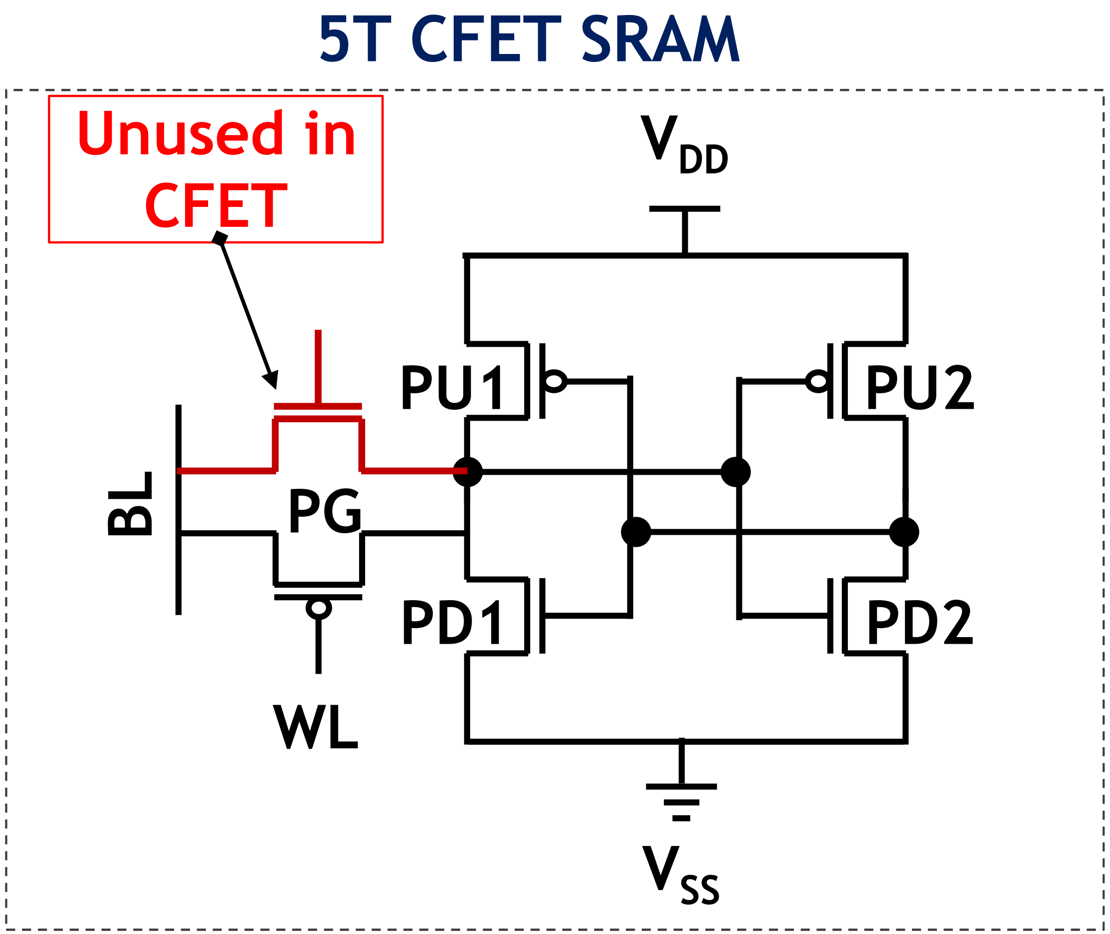
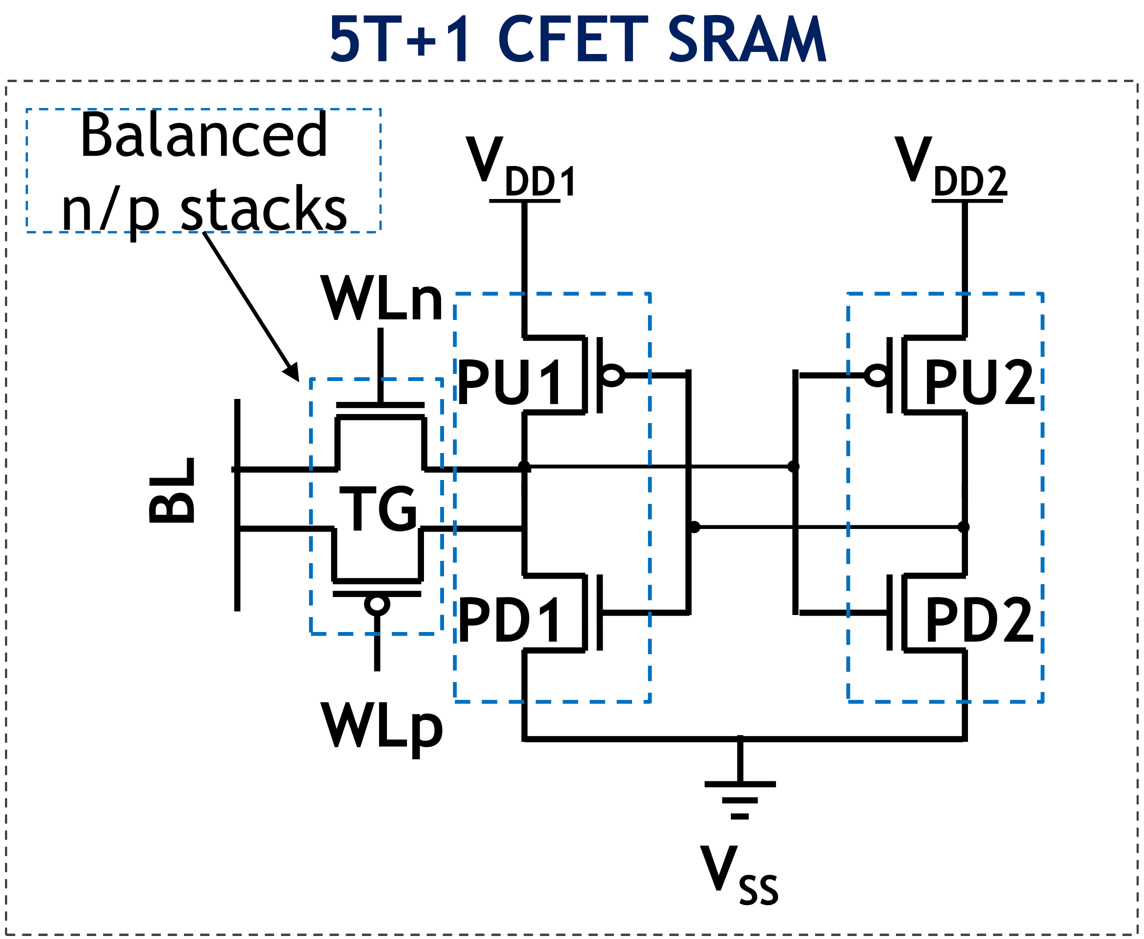
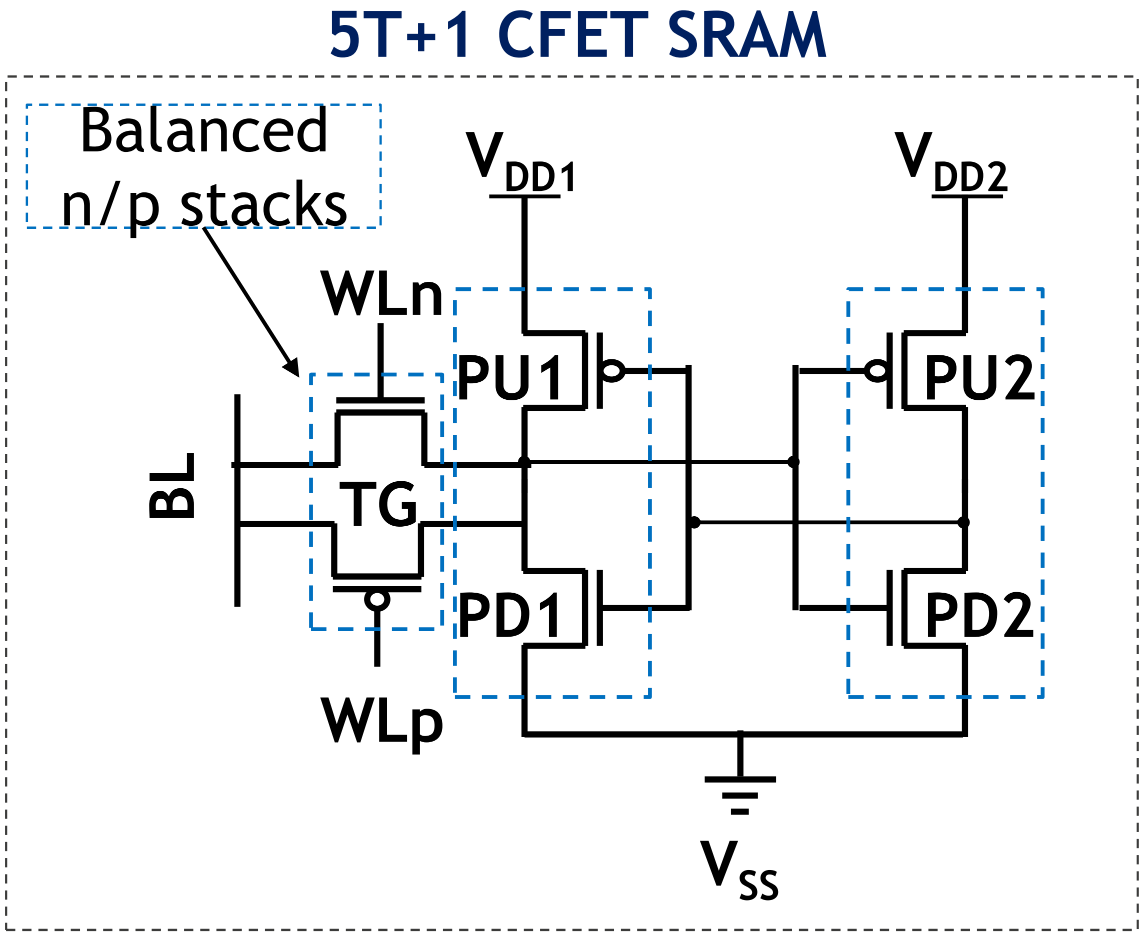
Proposed 5T+1 CFET SRAM bitcell

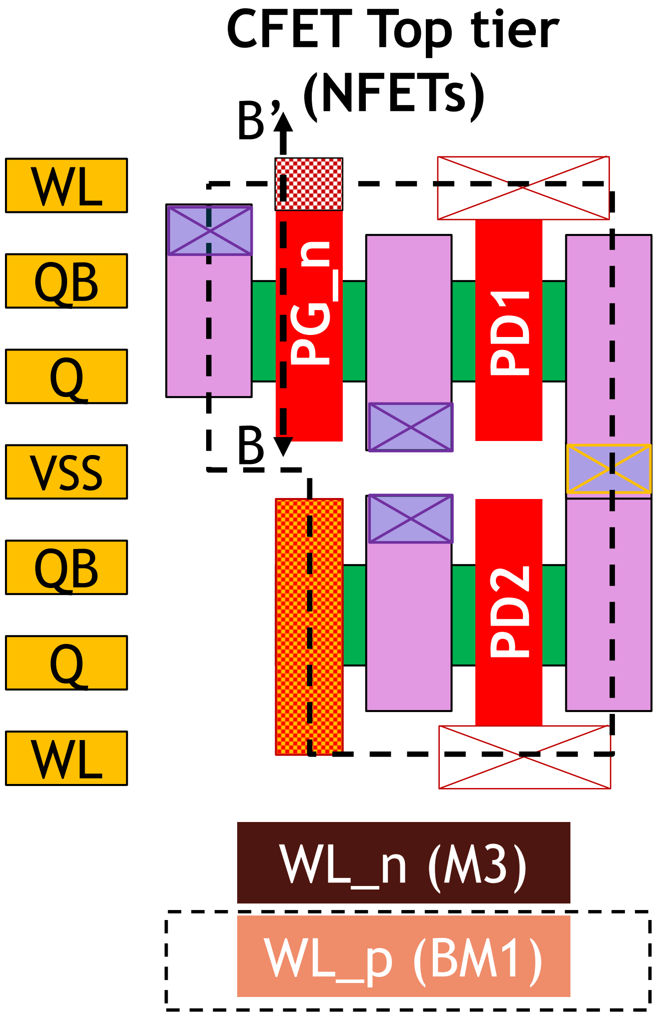


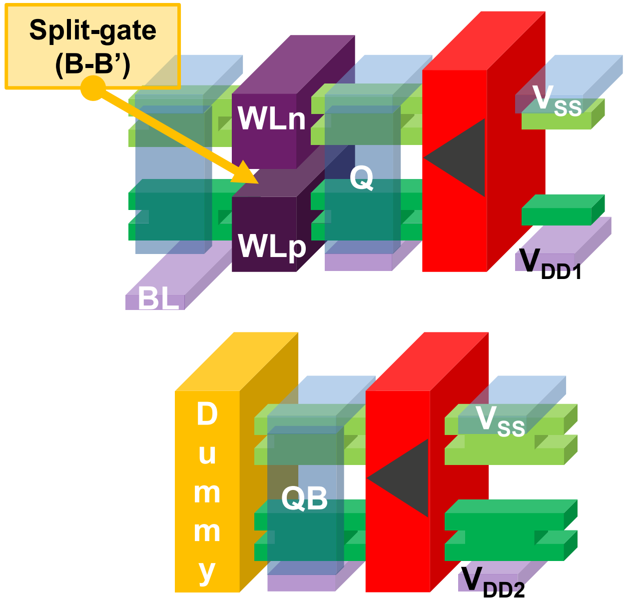


5T+1 CFET SRAM: Same footprint as 5T with additional access path → addresses 5T write/read limitations (next)

5T+1 CFET SRAM: layout implementation

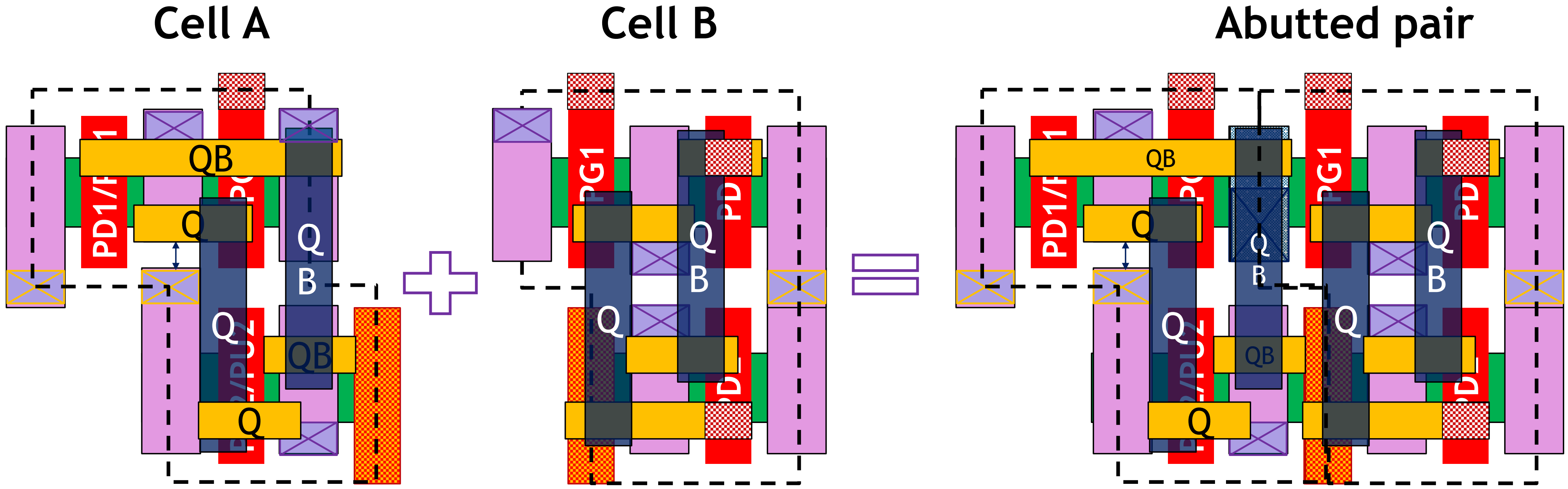






Eliminates nanosheet cut → requires split-gate + backside gate contact

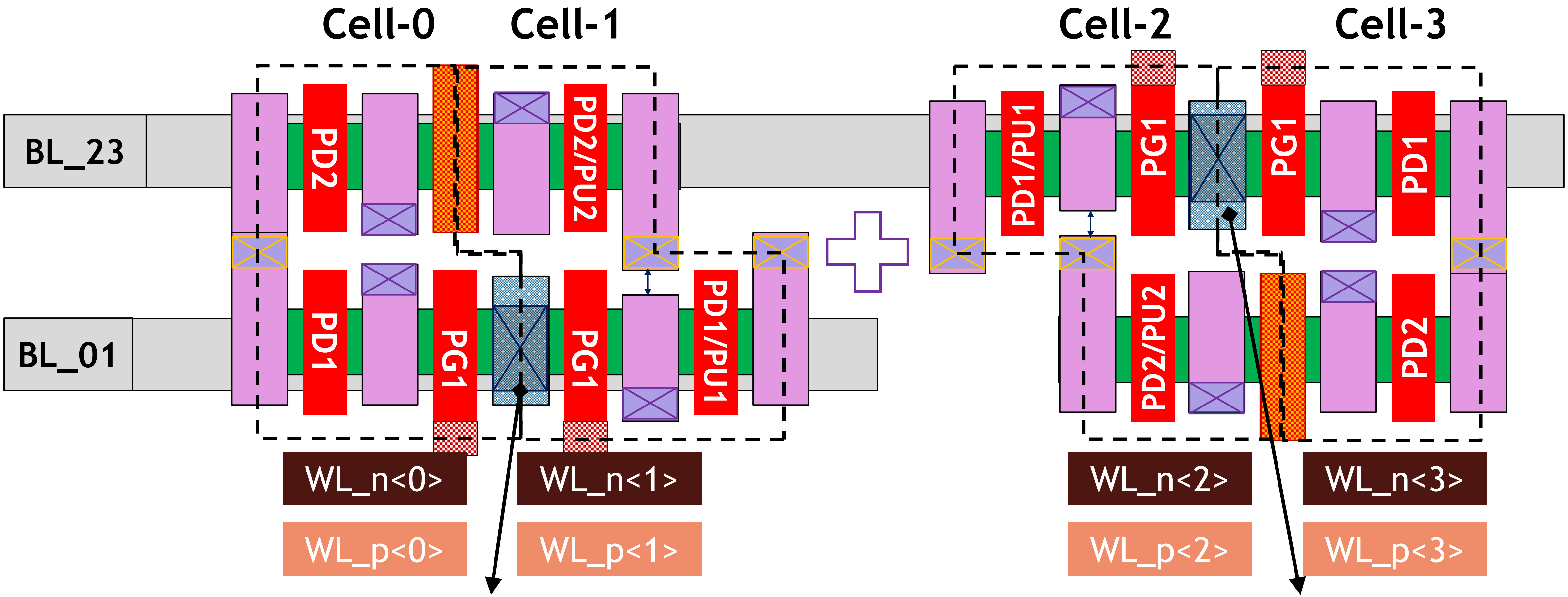
5T+1 CFET SRAM: complementary cells required



Non-rectangular footprint prevents uniform tiling

Complementary cell pairing enables tiling of non-rectangular cells

Dual-bitline-per-cell-height organization

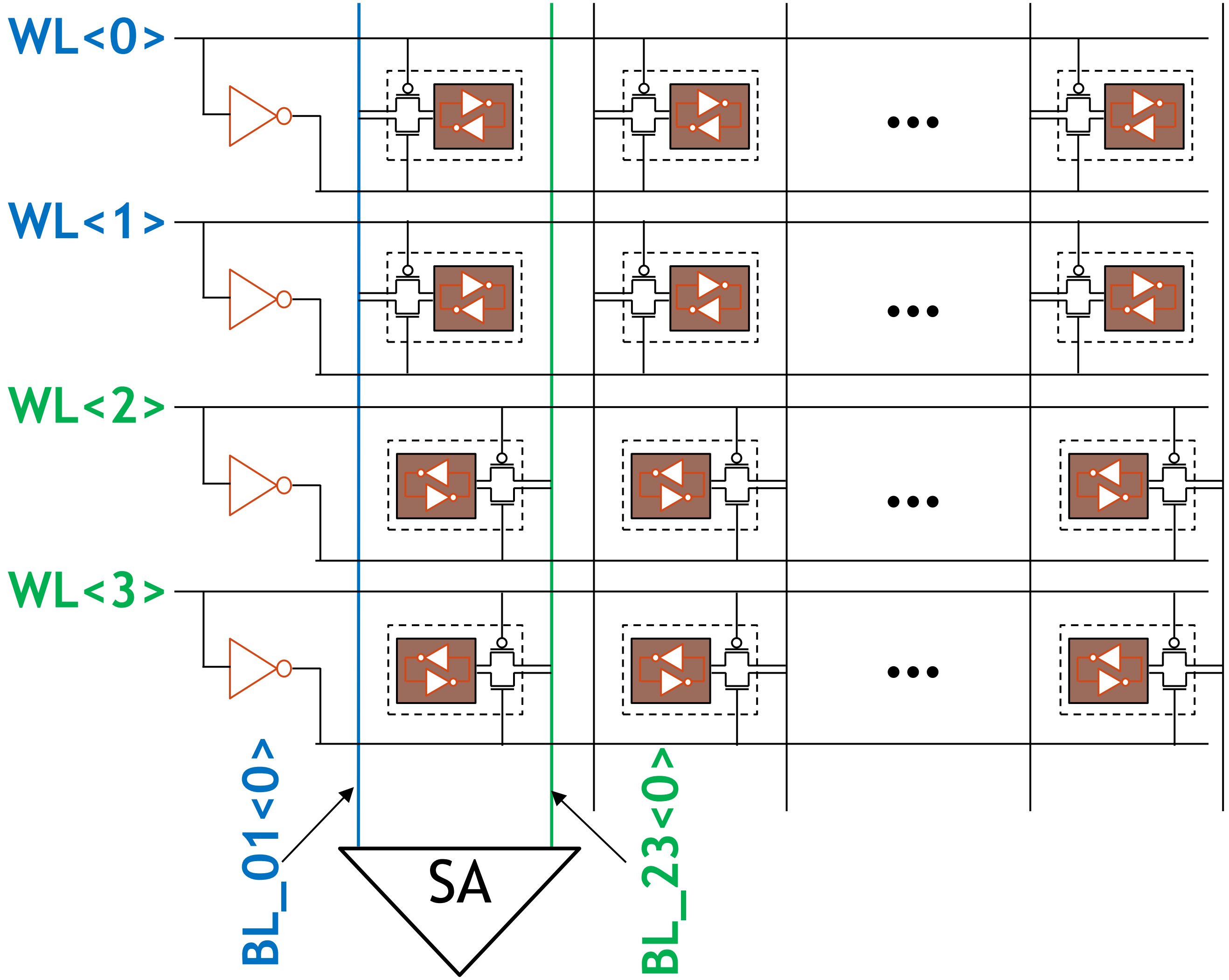


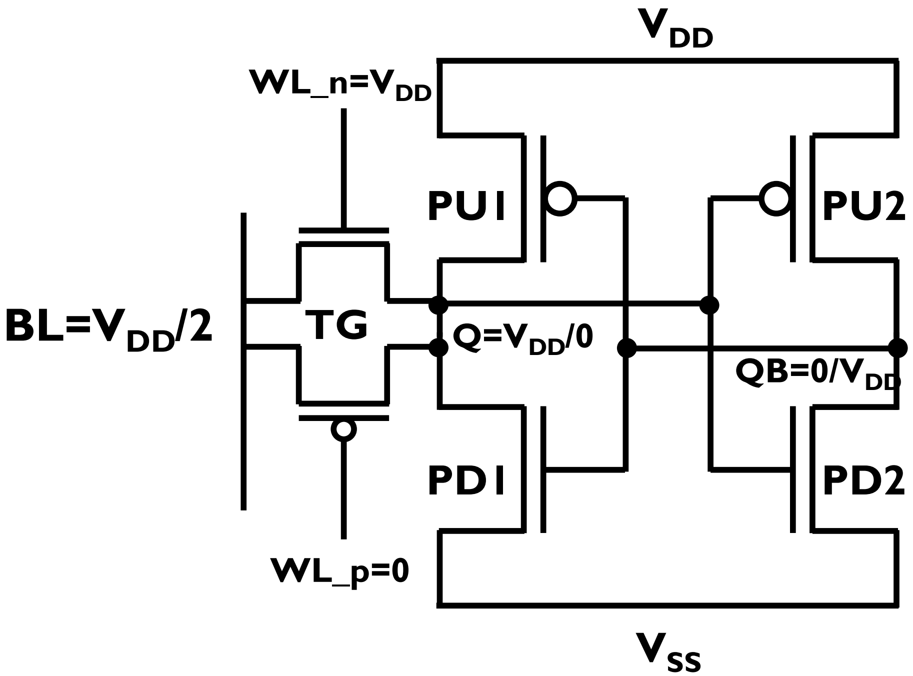
BL_01 (BVD+BM0) is shared b/n WL<0> and WL<1>

BL_23 (BVD+BM0) is shared b/n WL<2> and WL<3>

Differential sensing enabled via paired bitlines

Dual BL per column→one BL senses, the other serves as ref.→enables diff sensing

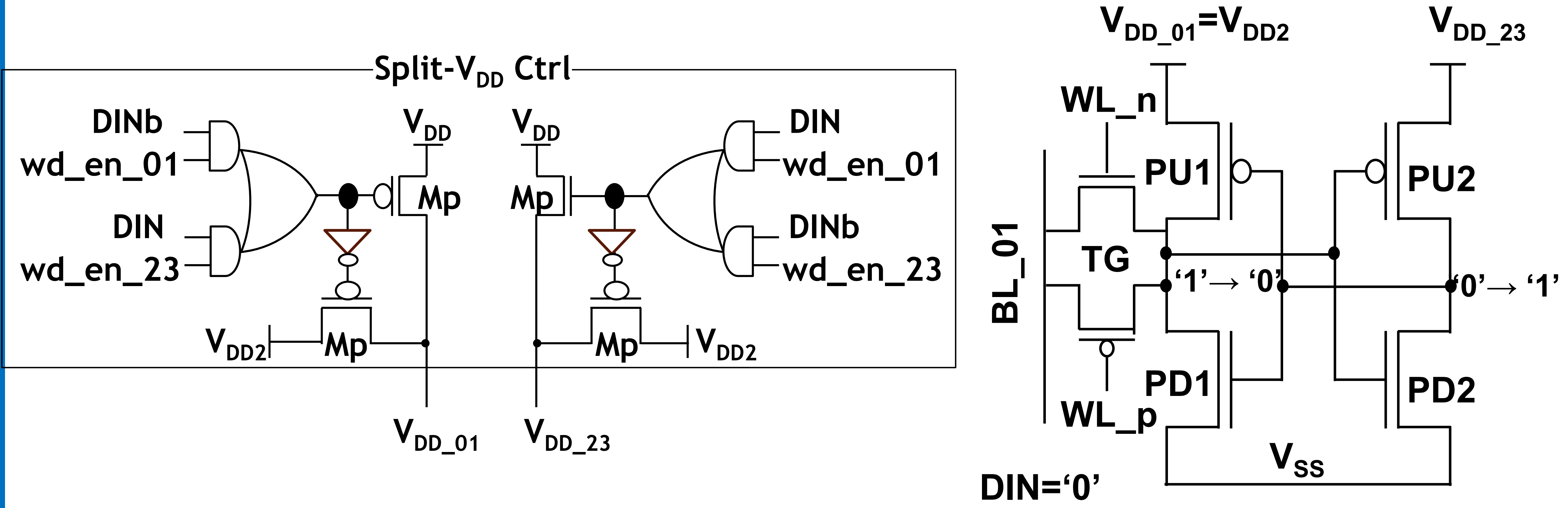




TG enables symmetric BL swing around ½ V_{DD} → also improves standby leakage

Split-VDD write assist

In-bitcell split-VDD write assist without cell footprint penalty



Column-selective VDD droop during write → weakens pull-up → enables robust write

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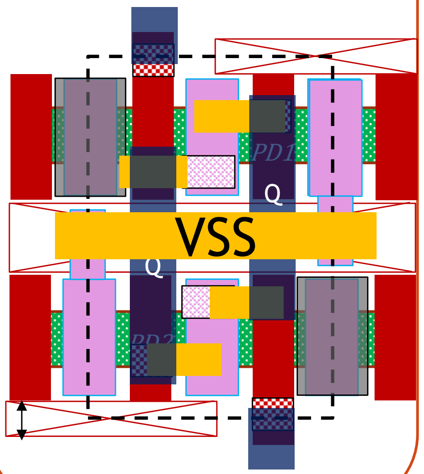
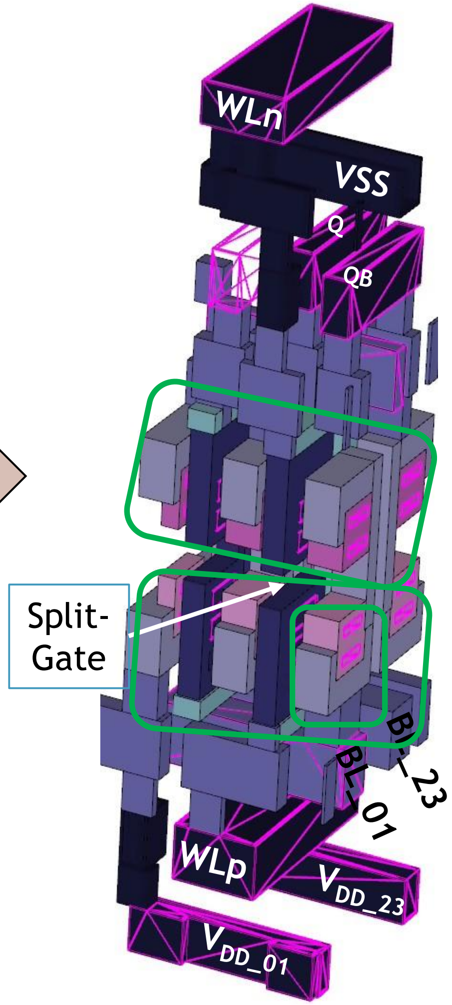
Layout-to-circuit DTCO evaluation flow

Design specifications	Technology specifications	TCAD simulations	SPICE simulations

SRAM bit-cell layouts





Parasitic extraction (PEX)

SRAM bit-cells pex netlist

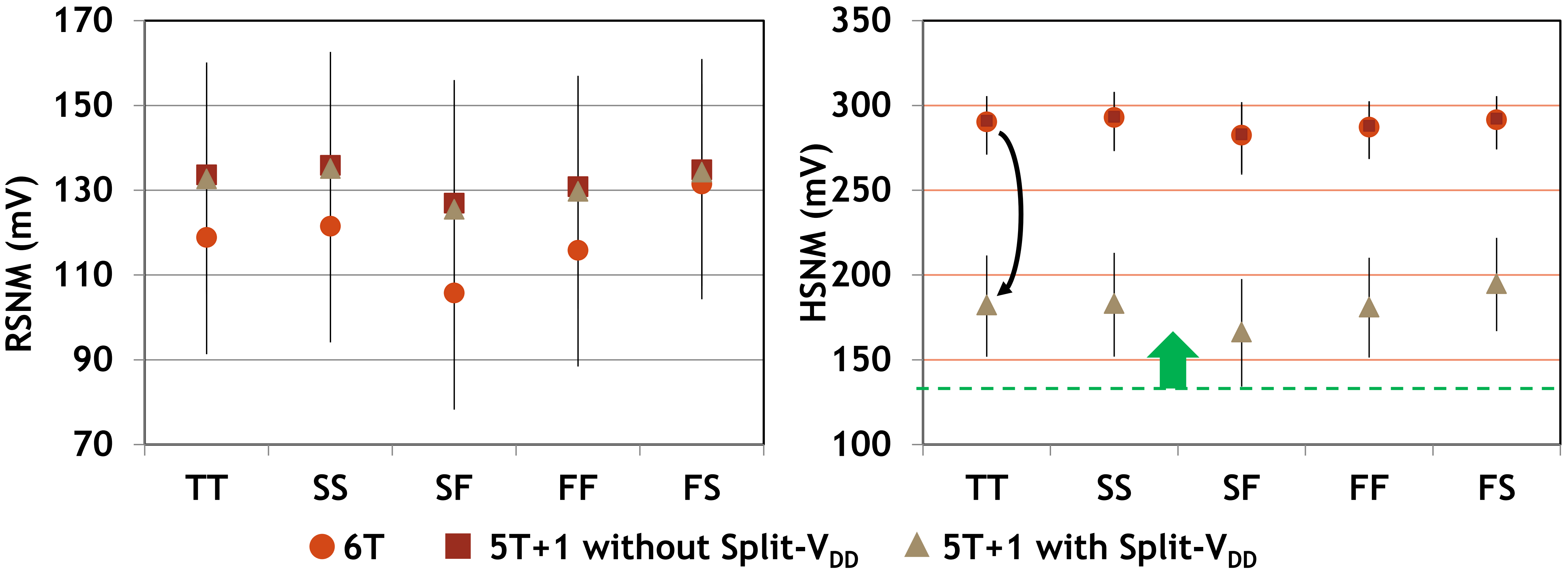
Device model cards

SRAM circuit tb

SRAMs KPIs
- Bit-cell level
- Array level

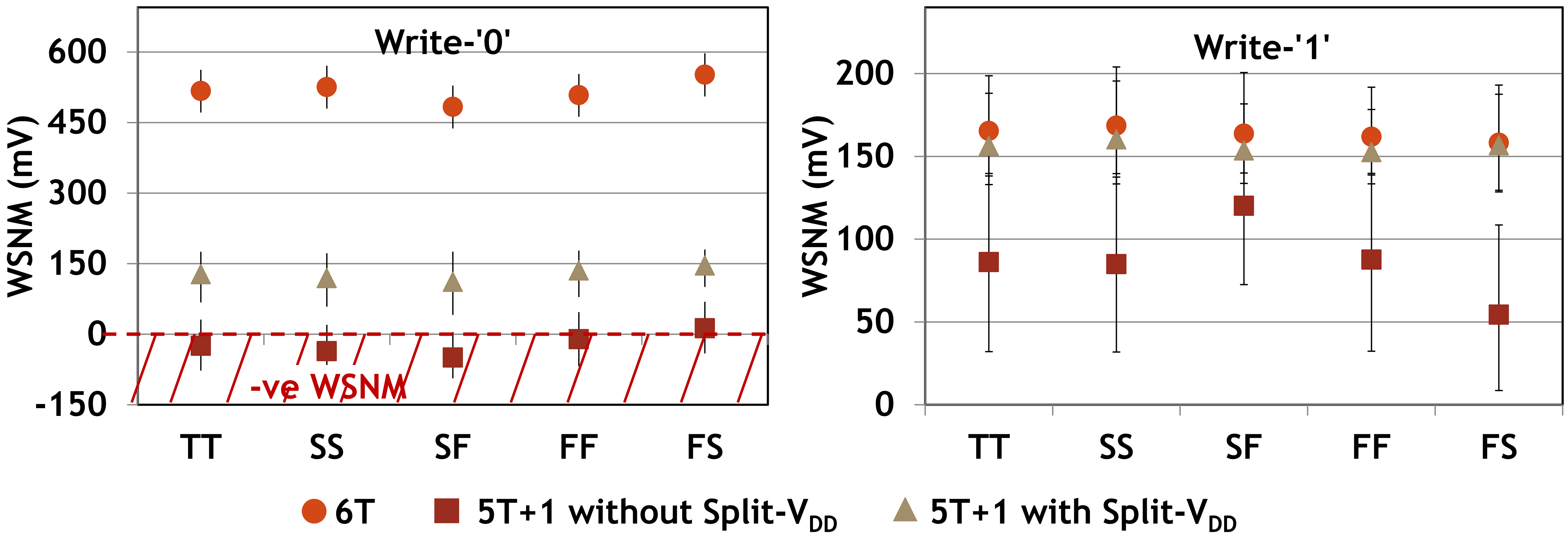
Adapted from: G.Rzepa, *et al.*, *IEDM*, IEEE (2022)

RSNM vs HSNM tradeoff in 5T+1 with split-VDD



No RSNM degradation; reduced HSNM in 5T+1 with split-VDD due to VDD droop in half-selected cells

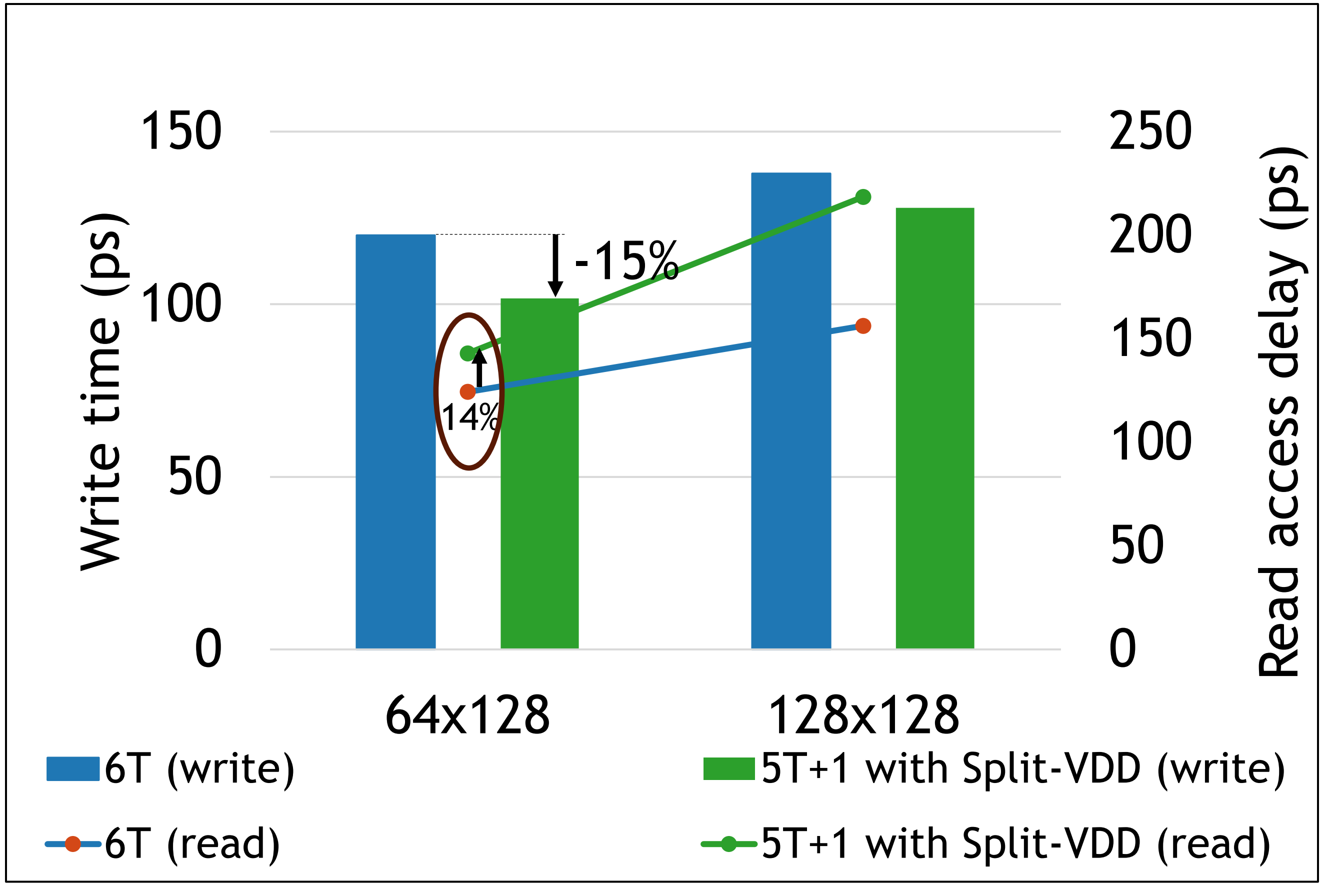
Write margin improvement with split-VDD



5T+1 without assist shows negative WSNM

Split-VDD restores write margin in 5T+1, eliminating negative WSNM

Write-read tradeoff and bitcell efficiency



Write improves (~15%) with minimal read penalty (~14%)

TABLE I Comparison with prior CFET SRAM work [1]

	VLSI'25 [1]	This work
Bitcell	6T CFET	5T+1 CFET
Bitcell area	1x	0.875x
Bitcell Leakage	1x	0.79x
Access device	PFET PG	TG access
Read sensing	Differ-ential (BL/BLB)	Different-ial (dual-BL/cell-height)
Write assist	-	split-V_{DD}
Split-gate	No	Yes
Top-NS cut	Yes	No
CFET stack utilization	Unbal-anced (4/2)	Balanced n/p stacks

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Conclusions

- 5T+1 CFET SRAM enables full utilization of CFET vertical stacking
  - **−12.5% area reduction vs 6T CFET SRAM**
- Dual-bitline-per-cell-height organization restores differential sensing
- In-bitcell split-V_{DD} write assist improves writability
- TG access enables symmetric $0.5 \cdot V_{DD}$ BL biasing sensing
- Modest read delay penalty with maintained SNM and reduced leakage
- **5T+1 CFET SRAM: viable high-density SRAM option for CFET nodes**

Thank you!